

MOHAK SHARMA

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EDUCATION

Columbia University – New York, NY

Dec 2026

Master of Science, Computer Science

Coursework: Machine Learning, Computational Neuroscience, Deep Learning, Computer Vision, Causal Inference, Quantum Computing, Storage Systems

Guru Gobind Singh Indraprastha University – Delhi, India

2019–2023

Bachelor of Technology, Computer Science — GPA: 9.29/10

PROJECTS & RESEARCH

CLARIS-Net: Neural-Driven Humanoid Avatar Control in Physics Simulation

BCI + RL + sim-to-real robot control

- Developed an end-to-end BCI system translating noisy EEG motor imagery into real-time, physics-grounded humanoid actions with < 100ms latency, aligned with Isaac Lab-style sim-to-real robot learning.
- Implemented PPO/SAC control in MuJoCo/PyBullet for a 17-DOF agent, refining decoded intent into natural bipedal motion and environment validation workflows.
- Leveraged EEGNet and contrastive learning for robust signal decoding, achieving > 90% accuracy in continuous joint-angle prediction and validating physical fidelity against motion-capture data.

Neural Radiance Fields – 3D Reconstruction

Neural rendering + camera-ray pipeline

- Engineered a coarse-to-fine NeRF with stratified and importance sampling over 5,000 training steps and 2,048 rays/step.
- Implemented the full camera-to-ray rendering stack from intrinsics matrix construction to world transforms; auto-selected best checkpoint by PSNR.
- Exported synchronized RGB and depth 60-frame sequences and integrated a Viser-based 3D visualization server for camera/ray/sample inspection.

MindSnap (Adaptive Intent Layer): Shared Autonomy for Multi-Modal Accessibility

POMDP-based shared autonomy

- Designed a Smart Intent Layer using POMDPs to model user goals as latent variables and Deep Q-Networks for shared autonomy.
- Built a cross-modal transformation engine for users with motor/sensory impairments, minimizing control effort via computational rationality frameworks.

CausalBench

Multi-agent causal evaluation

- Designed a multi-agent self-learning setup for causal evaluation, benchmarking GPT-4, Claude, and Gemini on confounding, selection bias, and counterfactual reasoning.

Mechanistic Interpretability of AlphaFold2

Representation analysis on 200K+ structures

- Applied UMAP-based dimensionality reduction and contrastive representation learning to 200K+ AlphaFold DB protein structures across 48 transformer layers.
- Revealed mechanistically interpretable substructures with implications for structure-based drug discovery and representation analysis.

WORK EXPERIENCE

Forward Deployed Fullstack Engineer – Growth & Product

Nov 2023 – Jul 2025

Outscal – Delhi, India

- Architected an AI-powered job parsing pipeline processing 2,000+ jobs/day (60K+ total), fine-tuning Gemini with 95%+ accuracy.
- Built custom GPT and Gemini models, recommendation systems, and A/B testing flows that drove a 15% conversion lift.
- Refactored the platform to TypeScript and improved SEO/performance, reducing critical-page Web Vitals to around 100ms.
- Led end-to-end creation of an internal no-code CMS platform, saving 60+ developer hours/week.

TECHNICAL SKILLS

Robotics & Simulation: MuJoCo, PyBullet, PPO, SAC, DQN, POMDP, reinforcement learning, imitation learning, sim-to-real transfer, domain randomization, URDF/MJCF

ML & Deep Learning: Python, PyTorch, TensorFlow, EEGNet, contrastive learning, CNNs, Transformers

Computer Vision & 3D: NeRFs, 3D reconstruction, COLMAP, Open3D, OpenCV, PyTorch3D, camera calibration, pose estimation

Systems & Cloud: AWS, FastAPI, Linux, Git, Redis, MongoDB, SQL, data pipelines

Full-Stack: TypeScript, JavaScript, React, Next.js, Redux Toolkit, GraphQL